Equations used by the ASM Lab solver

Version 2026.1.20 · SI units · P in Pa, Q in m³/s, mass flow in kg/s, h in kJ/kg, r in kg water per kg dry air, t in °C

Sources: the Solver Guide (labels A1–C11), the solver code (src/engine/solver, src/engine/thermal, src/lib/duct) and the items added since the guide, marked NEW. A tag in the first column gives the guide label where there is one.

1. Airway geometry and resistance

Ref.	Equation	Meaning
	$A = \frac{1}{2}\left\lvert \sum_i \left(x_i y_{i+1} - x_{i+1} y_i\right)\right\rvert, \qquad \mathrm{Per} = \sum_i \lvert p_{i+1} - p_i \rvert$	Cross-section area and perimeter from the outline (shoelace); a stated area and perimeter replace them
A1	$\Delta P = R\, Q\, \lvert Q \rvert$	Square-law pressure drop
A2	$R_{\mathrm{std}} = \dfrac{k\, L\, \mathrm{Per}}{A^3} + \dfrac{X\, \rho_{\mathrm{ref}}}{2A^2}$	Atkinson friction plus shock loss (X = shock factor)
	$R = R_{\mathrm{std}}\, \dfrac{\rho}{\rho_{\mathrm{ref}}} \quad (R \geq R_{\min} = 10^{-6})$	Resistance follows the air density; a typed R is locked
	$\rho_e = \frac{1}{2}(\rho_i + \rho_j)$	Airway density from its two node densities
A3	$R_{\mathrm{reg}} = \dfrac{1.44\, \rho}{A_r^2}$	Regulator of open area A_r (1.44 = 1.2², from ΔP = ρ(1.2Q/A)²)
	$R_{\mathrm{reg}} = R_{\mathrm{typed}}; \qquad A_r = 1.2\, \lvert Q \rvert \sqrt{\dfrac{\rho}{\Delta P_{\mathrm{target}}}}$ (target pressure drop, by bisection on Q); $\qquad Q = Q_{\mathrm{fixed}}$	Other regulator modes: fixed resistance, target pressure drop, fixed flow
NEW	$R_{\mathrm{wall}} = R_{\mathrm{catalogue}}$ or typed, $\qquad \Delta P = R_{\mathrm{wall}}\, Q \lvert Q \rvert$	Wall: brick 8500, concrete 25000, shotcrete 15000, good seal 80000, poor seal 2200, brattice 4000 Ns²/m⁸; the source tool presets 0 to 100000
NEW	$R_{\mathrm{orifice}} = \dfrac{\rho}{2\, C_d^2\, A_{\mathrm{free}}^2}, \quad C_d = 0.65, \quad A_{\mathrm{free}} = f_{\mathrm{open}}\, W\, H$	Louvre and rope choke from open area; a blocked wall is left out of the solve

2. Air state (psychrometrics)

Ref.	Equation	Meaning
A4	$e_s(t) = 610.5\, \exp\!\left(\dfrac{17.27\, t}{237.3 + t}\right)\ \mathrm{Pa}$	Saturation vapour pressure (Tetens; Blackbook 5.6)
A5	$p_w = e_s(t_{wb}) - \gamma\, P\, (t_{db} - t_{wb}), \qquad \gamma = 6.44 \times 10^{-4}\ \mathrm{K}^{-1}$	Vapour pressure from wet and dry bulb
	$r = \dfrac{0.622\, p_w}{P - p_w}, \qquad \mathrm{RH} = 100\, \dfrac{p_w}{e_s(t_{db})}$	Moisture content and relative humidity
	$t_{dp} = \dfrac{237.3\, \ln(p_w/610.5)}{17.27 - \ln(p_w/610.5)}$	Dew point
A6	$\rho = \dfrac{P - 0.378\, p_w}{287.04\, (t_{db} + 273.15)}$	Moist-air density (default)
NEW	$\rho = \dfrac{P - p_w}{287.04\, (t_{db} + 273.15)}$	Dry-air density basis (Settings → Engineering)
C2	$h = 1.005\, t_{db} + r\, (2501 + 1.8\, t_{db})$	Enthalpy of moist air
	$S = h - 0.03\, r\, L_0, \qquad L_0 = 2501\ \mathrm{kJ/kg}$	Sigma heat (the quantity conserved on mixing)
	$t_{db} = \dfrac{S - 0.97\, r\, L_0}{1.005 + 1.8\, r}$	Dry bulb from r and S; wet bulb then by Newton on A5
	$L_v(t) = 2501 - 2.383\, t\ \ \mathrm{kJ/kg}$	Latent heat at t

3. Natural ventilation and the network equations

Ref.	Equation	Meaning
A7	$\Delta_e = P_s - P_t - \rho_e\, g\,(z_t - z_s)$	Driving pressure of an airway including elevation
A8	$\mathrm{NVP} = g\,H\,(\bar{\rho}_{\mathrm{down}} - \bar{\rho}_{\mathrm{up}})$	Natural ventilation pressure
C1	$Q = \sqrt{\mathrm{NVP}/R_{\mathrm{loop}}}$	One-loop natural ventilation check
A9	$R\,q\lvert q\rvert - P_{\mathrm{fan}}(q) = \Delta_e$	Airway with a fan
A10	$F_i(P) = \sum_{e\to i}\rho_e q_e - \sum_{e\leftarrow i}\rho_e q_e = 0$	Mass balance at each free node (kg/s)
A11	$q_e = \mathrm{sign}(\Delta_e)\sqrt{\lvert\Delta_e\rvert/R_e}$	Flow from pressure difference
A12–13	$J(P^k)\,\delta P = -F(P^k), \qquad P^{k+1} = P^k + \alpha\,\delta P$	Newton–Raphson step
A14	$s_e = \dfrac{\partial q_e}{\partial \Delta_e} = \dfrac{1}{\lvert 2R_e\lvert q_e\rvert - dP_{\mathrm{fan}}/dq\rvert}$	Branch slope, capped near zero flow
A15	$J_{ii} = -\sum_{e\ni i}\rho_e s_e, \qquad J_{ij} = \sum_{e:\, i\leftrightarrow j}\rho_e s_e$	Jacobian assembly
A16	$\max_i \lvert F_i(P+\alpha\delta P)\rvert \le (1-c\alpha)\max_i\lvert F_i(P)\rvert, \quad c = 0.01$	Line-search (Armijo) condition
A17	$\mathrm{tol} = \max\!\left(10^{-4}\ \mathrm{kg/s},\ 10^{-6}\max_e\lvert\rho_e q_e\rvert\right)$	Convergence tolerance
B1–B4	$B\,x = c, \quad B = -J; \qquad l_{jj} = \sqrt{b_{jj} - \sum_{k<j} l_{jk}^2}, \quad l_{ij} = \dfrac{b_{ij} - \sum_{k<j} l_{ik}l_{jk}}{l_{jj}}; \qquad L\,y = c,\ L^{\mathsf{T}}x = y$	Direct (Cholesky) solve of each airflow step
B5	$\alpha_k = \dfrac{r_k^{\mathsf{T}} z_k}{p_k^{\mathsf{T}} B p_k},\ x_{k+1} = x_k + \alpha_k p_k,\ r_{k+1} = r_k - \alpha_k B p_k,\ \beta_k = \dfrac{r_{k+1}^{\mathsf{T}} z_{k+1}}{r_k^{\mathsf{T}} z_k}$	Preconditioned conjugate gradients (fall-back)

4. Fans

Ref.	Equation	Meaning
	$P(Q,\rho) = P_{\mathrm{ref}}(Q)\,\dfrac{\rho}{\rho_{\mathrm{ref}}}$	Fan law for density (can be switched off per fan or per project)
	$Q' = Q\,\dfrac{N'}{N}, \qquad P' = P\left(\dfrac{N'}{N}\right)^2, \qquad W' = W\left(\dfrac{N'}{N}\right)^3$	Affinity laws for speed or pulley ratio
	$W = \dfrac{Q\,P_{\mathrm{ref}}(Q)}{\eta(Q)}\cdot\dfrac{\rho}{\rho_{\mathrm{ref}}}$ (table with η); $\qquad W = W_{\mathrm{ref}}(Q)\,\dfrac{\rho}{\rho_{\mathrm{ref}}}$ (table with W)	Shaft power
	$P_{\mathrm{air}} = Q\,P, \qquad \eta = P_{\mathrm{air}}/W$	Air power and efficiency at the duty point
NEW	$\rho_{\mathrm{scale}} = \rho$ at the fan or $\rho_{\mathrm{design}} = \rho(P_{\mathrm{site}}, t_{db,\mathrm{site}}, t_{wb,\mathrm{site}})$	Setting: scale fans with the air at each fan, or with the surface design air (as the source tool)

5. Heat along an airway

Ref.	Equation	Meaning
C3	$h_{\mathrm{out}} = h_{\mathrm{in}} + \dfrac{g\,(z_{\mathrm{in}} - z_{\mathrm{out}})}{1000} + \dfrac{\dot{Q}_{\mathrm{added}}}{\dot{m}}$	Energy along an airway: autocompression and heat added
	$\dfrac{dt}{dz} = \dfrac{g}{1000\,(1.005 + 1.8\,r)}$	Dry autocompression lapse (about 9.7 °C per km)
C4	$\mathrm{VRT}(z) = \mathrm{VRT}_{\mathrm{datum}} + \dfrac{(z_{\mathrm{collar}} - z - d_{\mathrm{datum}})\cdot\mathrm{gradient}}{100}$	Virgin rock temperature with depth
C5	$q = U\,(\mathrm{VRT} - t_{\mathrm{air}}), \quad U = \dfrac{G(\mathrm{Fo},\mathrm{Bi})\,k_r}{r_0}, \quad \mathrm{Fo} = \dfrac{\alpha t}{r_0^2},\ \mathrm{Bi} = \dfrac{h_f r_0}{k_r},\ \alpha = \dfrac{k_r}{\rho_r c_r}$	Gibson/Starfield rock heat; t is the airway's age

Ref.	Equation	Meaning
C6	$\bar{G}(p) = \dfrac{\mathrm{Bi}\sqrt{p}\,K_1(\sqrt{p})}{p\big(\sqrt{p}\,K_1(\sqrt{p}) + \mathrm{Bi}\,K_0(\sqrt{p})\big)}$	Laplace-domain G, inverted numerically
	$h_f = \max\big(2,\ 0.71^{-2/3}\,k\,c_p\,v\big)\ \ \mathrm{W/m^2K}$	Air film coefficient from the airway's k and velocity, never below 2
C7	$\dot{Q}_s = \dot{m}\,c_p\,(\mathrm{VRT} - t_{db})\Big(1 - e^{-(1-w)\,U\,\mathrm{Per}\,L_s/(\dot{m}c_p)}\Big)$	Dry-wall sensible heat over a segment; w = wet fraction
NEW default	$U_r A(\mathrm{VRT} - t_s) = \dot{m}c_p(t_s - t)\,\varepsilon + L_v\,\dot{m}\,(r_{\mathrm{sat}}(t_s) - r)\,\varepsilon, \quad \varepsilon = 1 - e^{-hA/(\dot{m}c_p)}, \quad U_r = \dfrac{U\,h_f}{h_f - U}$	Water-film wet wall: the film temperature t_s is found by bisection; the wall cools the air towards its wet bulb
older	$q_l = \max\big(0,\ \min(w\,U\,A\,(\mathrm{VRT} - t_{wb}),\ \dot{m}(h_{\mathrm{sat}}(\mathrm{VRT}) - h) - q_s)\big), \quad \dot{m}_w = \dfrac{q_l}{L_v}$	Split model: the wet share of the rock heat evaporates; no air cooling
NEW default	$U = f\,U_{\mathrm{Gibson}}, \quad f = 1.8$	Rock heat factor, calibrated; no rock heat in a fan's own airway unless set. Wetness is a rating (Very dry 0, Dry 0.1, Damp 0.275, Wet 0.55, Very wet 0.85, default 0.1); the water-film model uses a tenth of it, wet = 0.1 × rating
	$q_{\mathrm{diesel}} = \dfrac{P_{\mathrm{rated}}\,\lambda}{\eta_e}\ (\eta_e = 0.33), \qquad \dot{m}_w = \dfrac{1.1\,q}{45000}$	Diesel heat (load factor λ) and combustion water
NEW	$q = \mathrm{kW} + \mathrm{kW/m} \cdot L, \qquad \dot{m}_w = \dfrac{\ell\,q}{L_v}$	Electric or other heat, with latent share ℓ that becomes water
	$q_{\mathrm{blast}} = \dfrac{m_{\mathrm{expl}} \cdot 3.7 \times 10^3 \cdot 0.2}{t_{\mathrm{flush}}}$	Explosives heat, flush time default 1800 s
	$q_{\mathrm{fan}} = W_{\mathrm{shaft}}$	All of the fan's shaft power goes into the air

6. Coolers

Ref.	Equation	Meaning
	$h_{\mathrm{out}} = h_{\mathrm{in}} - \dfrac{\mathrm{duty}}{\dot{m}}, \quad$ air saturated at h_{out}	Spray (direct-contact) cooler at a fixed duty
	towards saturated air at the coil's apparent dew point, 0.9 of the way	Coil (indirect) cooler; condenses below the dew point
	$t_{db} \to t_{db,\mathrm{target}},\ t_{wb} \to t_{wb,\mathrm{target}};$ never heating, optional capacity limit	Fixed-outlet cooler
NEW	$h_{\mathrm{out}} = h_{\mathrm{in}} - \varepsilon\,\big(h_{\mathrm{in}} - h_{\mathrm{sat}}(t_{w,\mathrm{in}})\big), \quad \mathrm{duty} = \dot{m}(h_{\mathrm{in}} - h_{\mathrm{out}}), \quad t_{w,\mathrm{out}} = t_{w,\mathrm{in}} + \dfrac{\mathrm{duty}}{\dot{m}_w \cdot 4.186}$	Bulk air cooler: effectiveness ε; the duty is held back so t_w,out ≤ the air's dry bulb on entry

7. Mixing and the coupled system

Ref.	Equation	Meaning
C8	$S_n = \dfrac{\sum \lvert\dot{m}_e\rvert\,S_{\mathrm{out},e} + M_{\mathrm{ext}}S_{\mathrm{ext}}}{\sum \lvert\dot{m}_e\rvert + M_{\mathrm{ext}}}, \qquad r_n =$ same with r	Mixing at a node (the guide writes it with h)
C9	$\ldots + \tau\,S_{\mathrm{still}}$ in the numerator, $+\tau$ in the denominator; $\quad \tau = \dfrac{\epsilon^2}{M + \epsilon}, \ \epsilon = 10^{-7}\max_e \lvert\dot{m}_e\rvert$	Node with almost no flow
C10	$F_n^{\mathrm{mass}} = \sum \rho q = 0, \quad F_n^{\mathrm{heat}} = h_n - h_{\mathrm{mix},n} = 0, \quad F_n^{\mathrm{moist}} = 1000\,\big(r_n - r_{\mathrm{fog}}(h_{\mathrm{mix}}, r_{\mathrm{mix}})\big) = 0$	Coupled mass, heat and moisture equations
C11	$J_{nk} = \begin{bmatrix} \partial F^m/\partial P & \partial F^m/\partial h & \partial F^m/\partial r \\ \partial F^h/\partial P & \partial F^h/\partial h & \partial F^h/\partial r \\ \partial F^r/\partial P & \partial F^r/\partial h & \partial F^r/\partial r \end{bmatrix}$	Block Jacobian, solved with GMRES (not symmetric)
B6	$v_1 = \dfrac{b}{\lVert b\rVert}, \quad w = Av_j, \quad h_{ij} = w^{\mathsf{T}}v_i, \quad w \leftarrow w - h_{ij}v_i, \quad v_{j+1} = \dfrac{w}{h_{j+1,j}}$	GMRES with Arnoldi

8. Ducts

Ref.	Equation	Meaning
	$R_{\text{duct}} = \dfrac{k_d\, L_{\text{eff}}\, \text{Per}}{A^3}, \quad k_d = 0.0055, \qquad L_{\text{eff}} = L + 0.041942\,\theta^{\circ}$	Duct friction with bends as equivalent length (θ in degrees)
	$L_{\text{red}} = \dfrac{k_c\, C_{\text{out}}\, L_{eq}}{A_{\text{out}}^3}, \quad k_c = 0.00316,\ L_{eq} = 1\ \text{m}$	Reducer loss, taken as a positive resistance
	$Q_{\text{leak}} = L_c\, \dfrac{\ell}{1000}\sqrt{\dfrac{\Delta P}{1000}}, \qquad R_{\text{leak}} = 1000\left(\dfrac{1000}{L_c\,\ell}\right)^2$	Duct leakage over a length ℓ (L_c in m³/s per km at 1 kPa)
	$u_c = u_0\, \min\left(1,\ \dfrac{B\,D}{s}\right),\ B = 5.8; \qquad u(r) = u_c\, e^{-r^2/b^2},\ b = 0.11\,s$	Free round jet from a duct mouth (flow preview only, not a solver result)
	$Q_{\text{face}} \geq 0.25\, A_{\text{tunnel}}\ \ (\text{m}^3/\text{s per m}^2)$	Minimum face quantity used as an indicator

9. Results derived after a solve

Ref.	Equation	Meaning
	$v = Q/A$	Velocity
NEW	$\text{share} = \dfrac{Q_{\text{working}}}{Q_{\text{intake}}}, \qquad \text{leakage} = \sum Q_{\text{walls}}$	Air use: air counted once through chains of working places
NEW	$\text{ventilation factor} = \dfrac{\dot{m}_{\text{intake}}}{\text{production (kt/day)}}\ \text{kg/s per kt/day}; \quad \text{per kt/month} = \dfrac{\text{per kt/day}}{30}$	Bluebook ch. 38: 3–8 (gold) and 2–5 (platinum) kg/s per kt/month
NEW	$H = 0.7\,\theta + \theta\sqrt{v}, \quad \theta = 36.5 - t_{wb}\ \ \text{mcal/cm}^2\,\text{s}, \qquad 1\ \text{mcal/cm}^2\,\text{s} = 4.186\ \text{mW/cm}^2$	Wet kata (Vent Calcs V16 workbook), from the air leaving each airway; the Blackbook asks for more than 12 in stopes and developments
NEW	$\text{SCP} = \text{SCP}_{\text{table}}(t_{wb}, v)\ \ \text{(bilinear)}, \qquad \text{SCP} \geq 300\ \text{W/m}^2$	Specific cooling power from the table in the Vent Calcs V16 workbook (wet bulb 20–34.9 °C, speed 0.1–4.5 m/s; 12 slips in it corrected). Light, moderate and hard work need 90, 180 and 270 W/m²
NEW	$t_{\text{re}} = \sum_{\text{path}} \dfrac{N\,V_e}{60\,Q_e}, \quad N = 8,\ V_e = A_e L_e$	Re-entry time after blasting: air changes N of each airway's volume at its solved flow, summed along the longest path the fumes take; at least 30 min for a multi-blast

Specific cooling power is read from a table, not from an equation: neither the Blackbook nor the Bluebook gives one (both cite Stewart 1989), and the Vent Calcs V16 workbook looks it up by wet bulb and speed.